DS	**Session**	Graph	**Question Information**	Level 1 Challenge 84

Question description

Byteland has n cities and m roads between them. The goal is to construct new roads so that there is a route between any two cities.
Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Constraints

$1 \le n \le 10^5$
$1 \le m \le 2 \cdot 10^5$
$1 \le a,b \le n$

Input

The first input line has two integers n and m: the number of cities and roads. The cities are numbered 1,2,...,n.

After that, there are m lines describing the roads. Each line has two integers a and b: there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k: the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4 2
1 2
3 4
```

EXPECTED OUTPUT

```
1
2 4
```

Test Case 2

INPUT (STDIN)

```
4 2
2 3
2 4
```

EXPECTED OUTPUT

```
1
1 4
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(m--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
5
```

Test Case 2

TOKEN COUNT

```
475
```

Test Case 3

NLOC

```
50
```

Code

C building_roads.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXN 100005

int parent[MAXN];

int find_rep(int v) {
    if (v == parent[v]) return v;
    return parent[v] = find_rep(parent[v]);
}

void union_sets(int a, int b) {
    a = find_rep(a);
    b = find_rep(b);
    if (a != b) parent[b] = a;
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

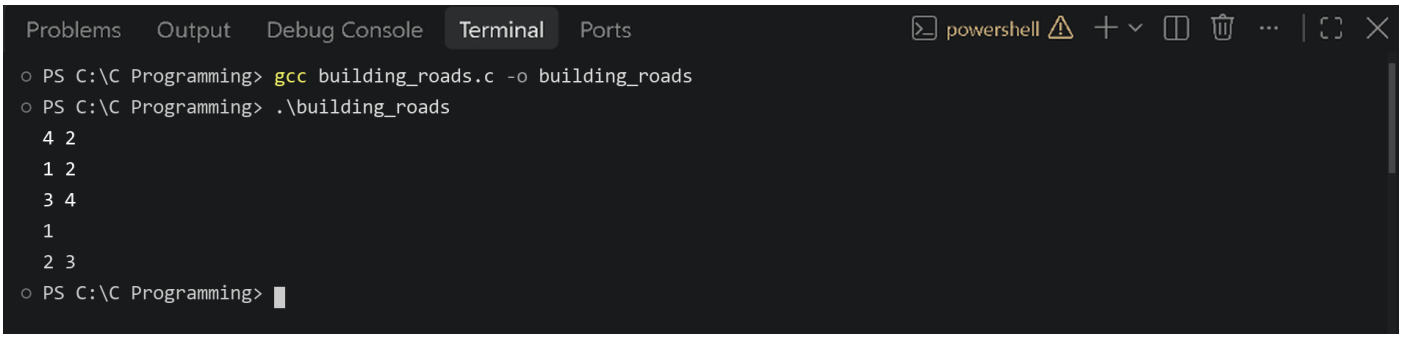
    for (int i = 1; i <= n; i++) parent[i] = i;

    while(m--) {
        int u, v;
        scanf("%d %d", &u, &v);
        union_sets(u, v);
    }

    int reps[MAXN];
    int comp_count = 0;
    for (int i = 1; i <= n; i++) {
        if (find_rep(i) == i) {
            reps[comp_count++] = i;
        }
    }

    int k = comp_count - 1;
    printf("%d\n", k);
    for (int i = 0; i < k; i++) {
        printf("%d %d\n", reps[i], reps[i + 1]);
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc building_roads.c -o building_roads
PS C:\C Programming> .\building_roads
4 2
1 2
3 4
1
2 3
PS C:\C Programming>